

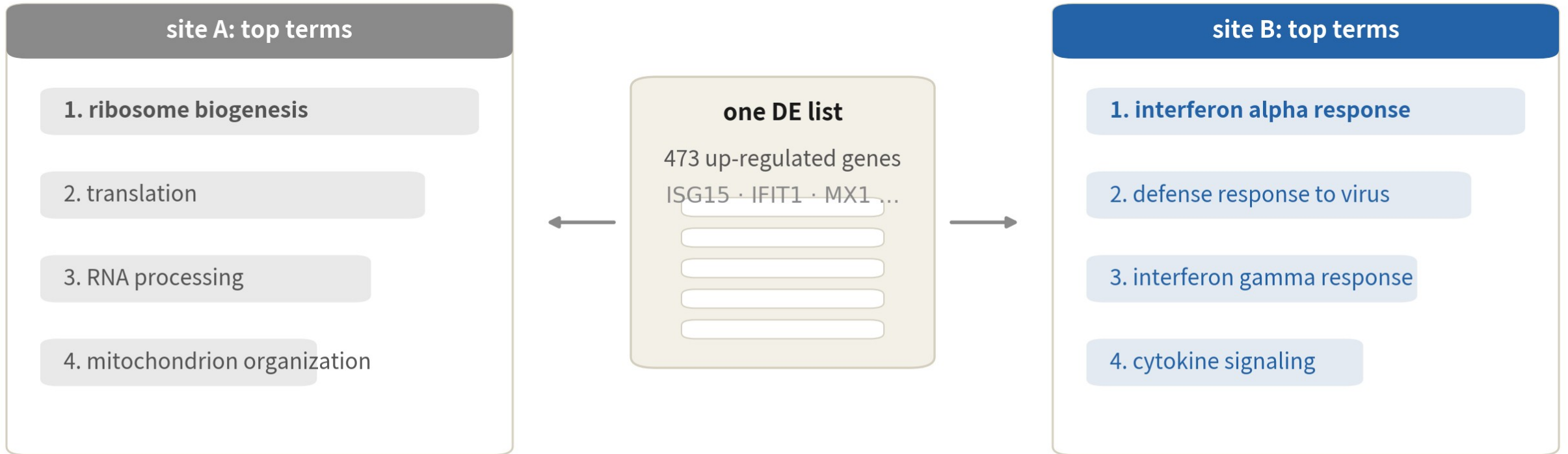
BS5 Series B Supplement

A field guide to enrichment resources: from gene list to biology

Four iron rules, a database map, a toolchain map, one decision tree — from 'it runs' to 'it's the right choice'.

About 35 min | Prerequisites: B12, B13 | Series B Supplement, Ep. 5

One list, two websites, two answers



Same list, submitted the same day — the two top boards barely overlap. Who is lying?

schematic

Housekeeping themes on the left, interferon everywhere on the right — the same list

WHERE THIS EPISODE STARTS

Nobody is lying: the two sites used different universes.

Site A used the whole genome as background; site B used the detected genes. Both tools are correct — the choices differ. This episode is about those choices.

What you will learn

- 1 Use four iron rules to fix the method, the background, and the version record before you run anything
- 2 Read the gene-set database map and its licence traffic light — including KEGG's triple restriction
- 3 Walk the decision tree and pick the right toolchain for a list, a ranking, per-cell scores, or activity

01

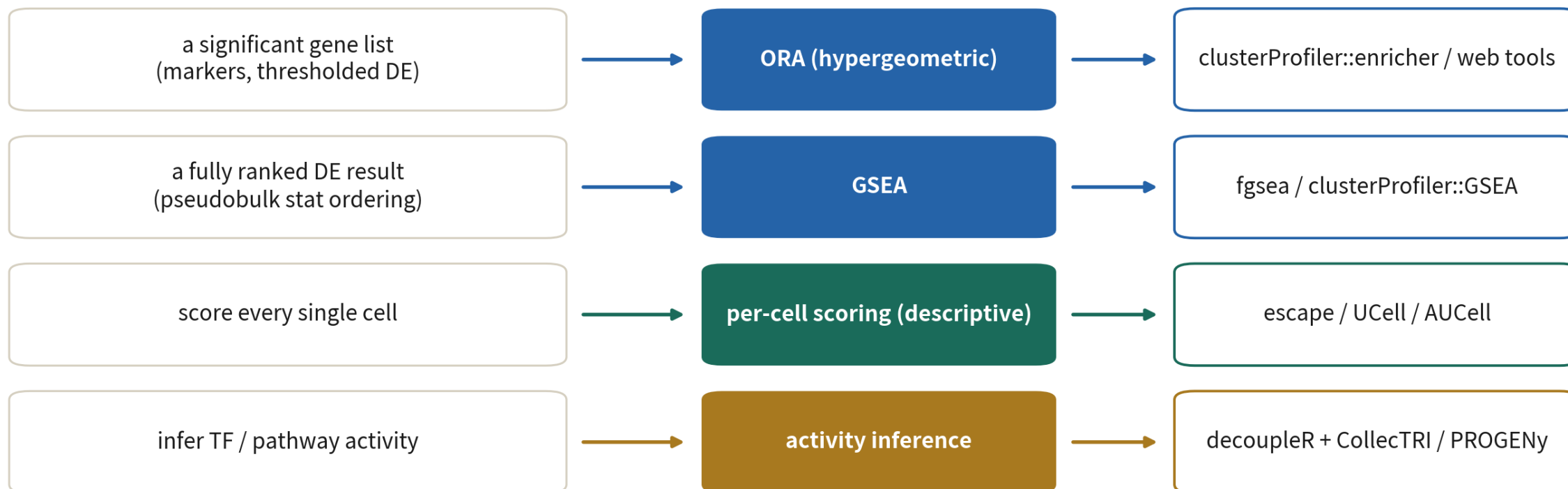
The four iron rules

The mechanics of GSEA and ORA live in B13; here we set the principles of choice

Rule 1: the input decides the method

What do you have?

schematic



Common mismatch: thresholding by p and then running GSEA — you threw the ranking away yourself

Ask what you hold first — the method and the tool fall into place

Lists take ORA; rankings take GSEA

B13's conclusion, reused as the first fork

ORA

You only have a list

- input: a thresholded list plus a background
- hypergeometric test; cutoff-sensitive
- a wrong background poisons everything (demo coming)

GSEA

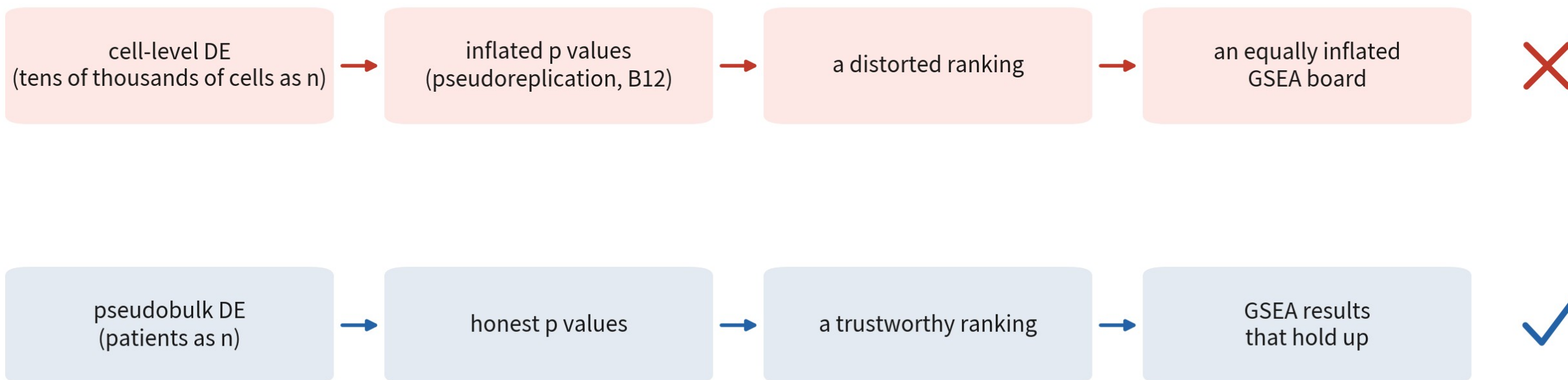
You have the full ranking

- input: the whole ordering, no cutoff
- fgsea / clusterProfiler::GSEA
- thresholding before GSEA throws information away

Rule 2: chase the unit of analysis upstream

the same multi-patient dataset

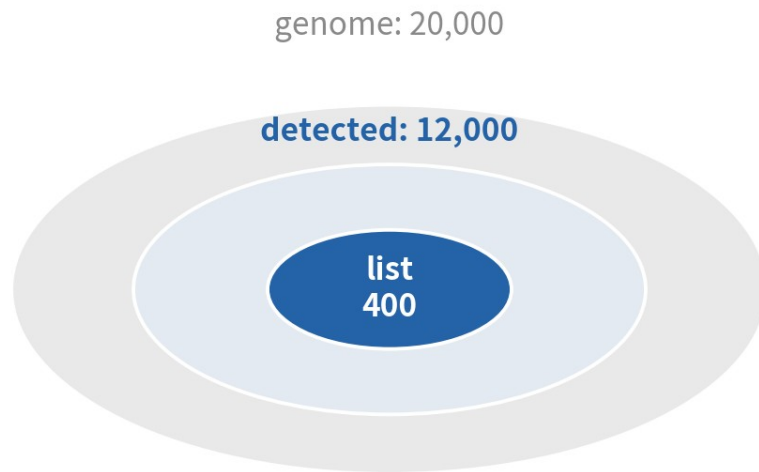
schematic



Enrichment never rescues an upstream mistake — garbage ranking in, garbage board out

An enrichment result is only as trustworthy as the DE ranking you fed it (B12)

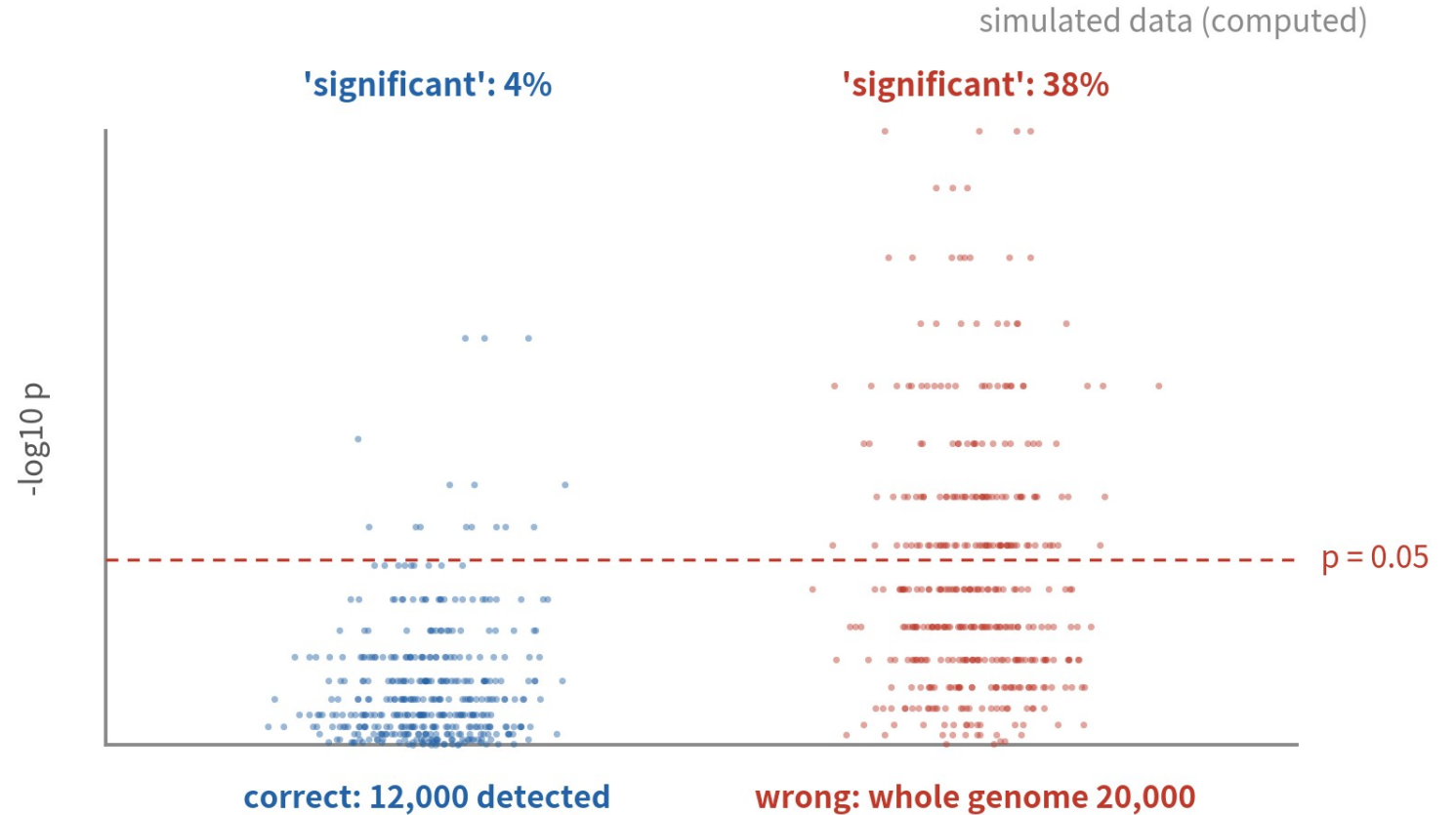
Rule 3: the universe is the invisible error



the list can only come from the middle ring

The list is random — zero true enrichment; the wrong background reads 'detectable' as 'enriched'

A random list with zero true enrichment — the wrong background still yields 38% 'significant'



Writing the universe correctly

ORA has two inputs, not one: the list and the background

```
sig.up <- rownames(pb)[pb$padj < 0.05 & pb$log2FoldChange > 1]
universe <- rownames(pb) # 背景 = 這次實際檢定過的基因
length(sig.up); length(universe)

library(msigdb); library(clusterProfiler)
hm <- msigdb(species = "Homo sapiens", collection = "H")
ora <- enricher(sig.up,
  universe = universe, # 不設，工具拿全基因組頂上
  TERM2GENE = hm[, c("gs_name", "gene_symbol")])
```

```
[1] 473
[1] 11143
```

The B13 ifnb example: 473 in the list, 11,143 in the background — both numbers go into the report

Rule 4: record the versions — three things

The enrichment edition of 'a number, a reason, a record' — miss one and the result is unreproducible

Which database, which version



MSigDB 2026.1?
Which GO release?
Databases move every
year — and the board
moves with them

Which method



ORA or GSEA? Which
scorer? Whose
defaults did the
parameters come
from?

Which background



What the universe was
and how many genes it
held — noted down
like a QC threshold

CORE IDEA

**Answer the four rules first;
only then pick databases and tools.**

Input decides method; chase the unit upstream; set the universe; record versions. Answer these and the rest is table lookup.

02

The gene-set databases

MSigDB, GO, Reactome, WikiPathways — and KEGG's licensing landmine

MSigDB 2026.1: the one-stop warehouse

MSigDB v2026.1 — 35,361 gene sets

H 50 Hallmark low-redundancy opener	C1 302 positional chromosome bands	C2 7,670 curated incl. Reactome / WP / legacy KEGG	C3 3,714 regulatory targets TF / miRNA targets	C4 1,006 computational computationally derived
C5 16,283 ontology GO + HPO	C6 189 oncogenic oncogenic signatures	C7 5,219 immunologic immunologic signatures	C8 866 cell type scRNA-seq derived	C9 62 perturbation new in 2026

Access: website download needs free registration; the msgdbr R package needs none (multi-species, tidy)

figures from official pages (verified 2026-09)

35,361 gene sets across ten collections — you only need to know a handful

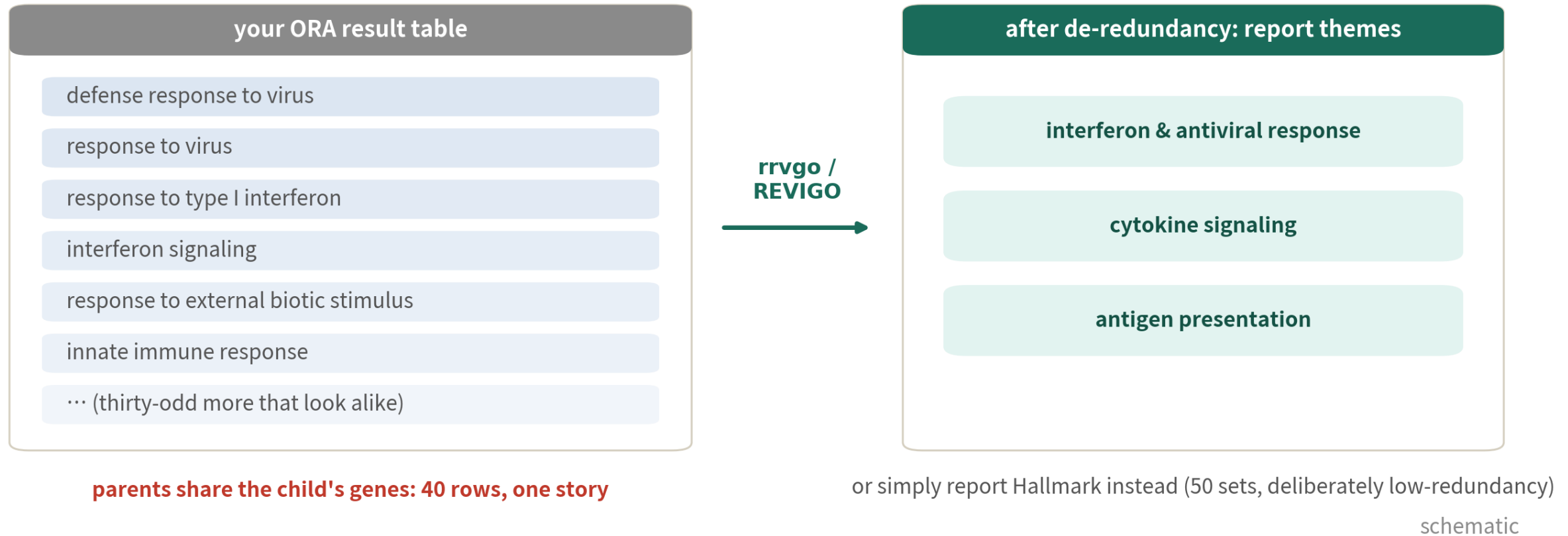
The collections you will actually use

2026.1 adds C9 — anywhere your notes say 'C1–C8' now needs updating

Collection	Size	When to use it
H • Hallmark	50	always the opening move: low-redundancy, easy to report
C2 • curated	7,670	holds the Reactome / WikiPathways / legacy KEGG subsets
C5 • ontology	16,283	GO + HPO; highly redundant — pair with de-redundancy tools
C7 • immunologic	5,219	immunologic perturbation signatures for immunology projects
C8 • cell type	866	cell-type marker sets derived from scRNA-seq
C9 • perturbation	62 (new)	computational perturbation signatures (opened 2026)

GO: the biggest — and the most redundant

GO release 2026-08-05: 38,092 terms (BP 23,974 / MF 10,041 / CC 4,077)



BP is the default for marker-list ORA; when hundreds of terms return, de-redundify before reporting

Reactome vs WikiPathways: two CC0 pathway libraries

The free pair for mechanistic reading — both commercial-safe

Reactome

Hierarchical, expert-curated

- V97: 2,883 human pathways
- data in CC0, free, no registration
- first pick for signalling / metabolism — the legal KEGG substitute

WikiPathways

Community-curated, fast-moving

- 2,090 pathways (2026-08 release)
- CC0; native clusterProfiler support (enrichWP)
- new topics the big two haven't curated often land here

KEGG: understand the licence before you use it

KEGG (release 119.0 • 587 reference pathway maps)

1 bulk FTP download

subscription only —
academia pays too

2 REST API

academic users only,
 ≤ 3 requests / second

3 commercial use

requires a paid licence

clusterProfiler's enrichKEGG / gseKEGG fetch KEGG online at run time — fine in academia, not in industry

Legal alternatives: Reactome (CC0) / WikiPathways (CC0) / the legacy KEGG sets inside MSigDB C2

licence terms from KEGG's official legal page (verified 2026-09)

The most important landmine card of this episode — especially if industry is in your future

KEGG: when to use it, when to route around

Why it survives



its pathway diagrams are still the best in class (pathview colouring); fine for academic teaching

Why you route around



commercial projects and offline bulk gene-set pipelines — the licence blocks you

The detour



Reactome /
WikiPathways (CC0), or
the legacy KEGG sets
inside MSigDB C2

The licence traffic light

figures from official pages (verified 2026-09)

green: safe even commercially

Reactome	CC0 (public domain) • V97, 2,883 human pathways
WikiPathways	CC0 • 2,090 pathways, community-curated
STRING	CC BY 4.0 (attribute and go) • v12.5
Gene Ontology	free, no registration • release 2026-08-05

amber: usable, with a gate

MSigDB	site download needs free sign-up; msigdb needs none
DISGENET	commercialised: free tier by academic application; no more bulk download

red: route around it commercially

KEGG	triple restriction: FTP subscription / academic-only API / paid commercial licence
viper	non-commercial licence (one of decoupleR's methods)

Nearly everything is green in academia; in industry, check this chart before choosing

LANDMINE WARNING

**In a commercial setting,
licensing is a legal problem, not a technical
one.**

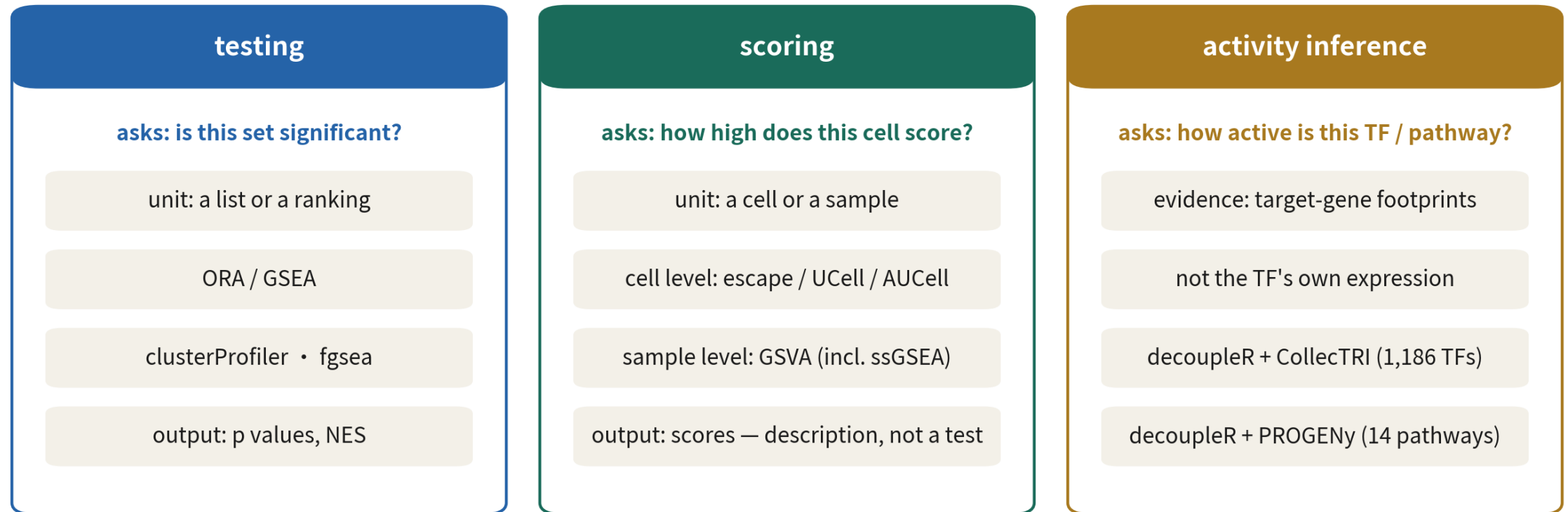
enrichKEGG runs just fine on a company laptop — running is not the same as being allowed to run.

03

The toolchains

Testing, scoring, activity inference — three routes, each with its workhorse

Three routes, three questions



All three routes report at the cluster / patient level — chase the unit of analysis upstream (B12)

schematic

Fix which question you are asking, then pick tools from that route

The testing route's three-piece set

B13's teaching stack — all on Bioconductor

Package	Version	Role
clusterProfiler	v4.20.0	one interface for ORA + GSEA; takes GO / WikiPathways / any GMT
fgsea	v1.38.0	preranked GSEA; fast — built for batches of contrasts
enrichplot	v1.32.0	the dotplot / cnetplot / emapplot / gseaplot2 production line

The scoring route: GSVA vs per-cell

Both are scoring — different levels, different tools

per sample

GSVA (v2.6.6)

- gene-by-sample to geneset-by-sample; four methods incl. ssgsea
- pathway scores on pseudobulk, then between-group stats
- runs at cell level, but dropout bites

per cell

escape / UCell / AUCell

- escape (v2.8.0), the one-stop shop: four scorers, one interface
- UCell / AUCell: rank-based, steadier under dropout
- scores describe; conclusions return to cluster / patient level

The activity-inference kit

Activity is not expression — TF mRNA is lowly expressed and dropout-ridden

decoupleR



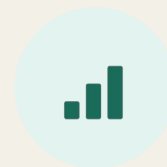
one interface, many inference methods; R v2.17.0 — the official focus has moved to Python's decoupler

CollecTRI: the TF dictionary



1,186 TFs and 43,175 signed TF-gene links; DoRothEA's official successor

PROGENy: pathway footprints



14 pathways; activity inferred from perturbation-derived footprint genes

Maintenance snapshot (verified 2026-09)

Tools age — so will this table; recheck the official pages before citing

Resource	Status	What to do
DoRothEA	officially superseded by CollecTRI	use only to reproduce pre-2023 papers
SCENIC (R)	officially deprecated	use pySCENIC; evaluate SCENIC+ for new projects
pySCENIC	latest release 0.12.1 (2022-11), slow-moving	state the maintenance status in teaching and reports
viper	non-commercial licence	switch to ulm / mlm and friends commercially

The three quick-look websites

Paste a list, get an overview in thirty seconds — scout first, formalise later

Tool	Scale / version	Strength
Enrichr	225 gene-set libraries; no login	fastest cross-library overview; the classroom favourite
g:Profiler	Ensembl 114, 400+ species	the non-model-organism rescue; gprofiler2 plugs into R
ShinyGO	v0.86 (Ensembl 115, 925 species)	ready-made figures, no login
DAVID	Knowledgebase v2025_2; older snapshot	for cross-checking older papers only

The web tools' three limits

No universe control



the site chooses the background — exactly how the cold open happened

No version control



the day the site updates its libraries, your result stops reproducing

Formal results go back to R



clusterProfiler + msigdb + recorded versions — that is the road to a paper

TAKE THIS LINE

Web tools are reconnaissance, not the report.

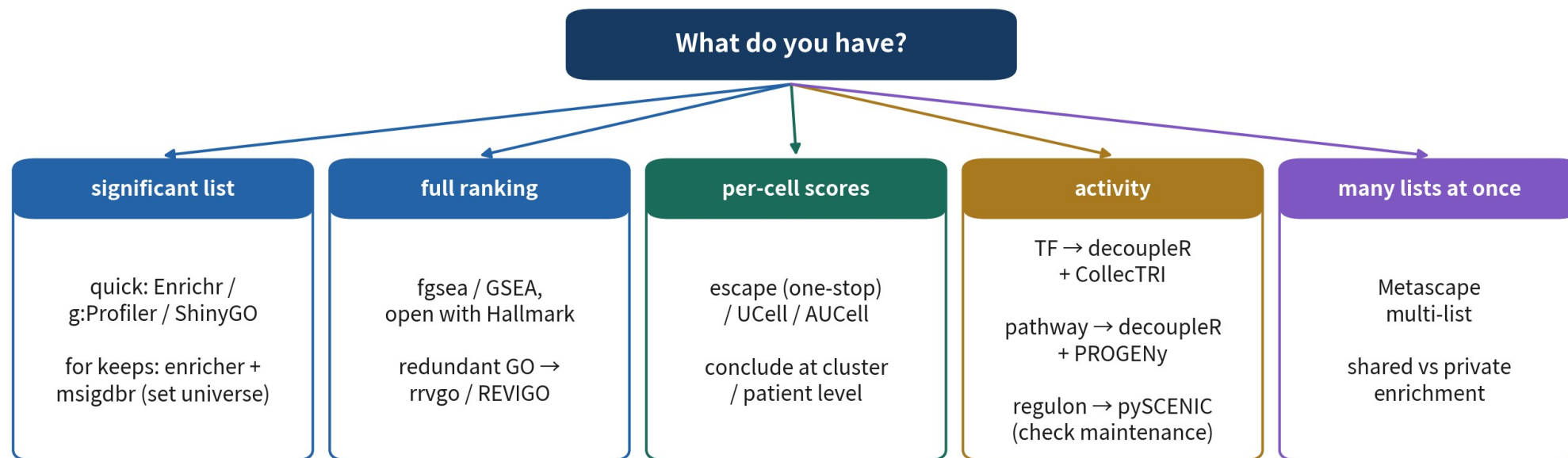
Reconnaissance buys direction; every number that enters the paper walks the reproducible,
versioned road.

04

Walking the decision tree

One tree, five branches, two worked scenarios

The decision tree: start from what you hold



Which database?

- always open with: MSigDB Hallmark (50) + GO BP
- signalling / metabolism → Reactome (CC0) > KEGG (triple licence restriction)
- new topics, community-curated → WikiPathways (CC0)
- commercial settings → avoid KEGG; use Reactome / WikiPathways / STRING (CC BY)

full tree: 富集分析參考庫/總覽說明.md

This episode's signature figure — the full version lives in the enrichment reference library

Five branches in one line each

You hold	Route	Remember
a significant list	web tools to scout; enricher + msigdb for keeps	set the universe (Rule 3)
a full ranking	fgsea, with Hallmark as the first gene-set choice	rank from pseudobulk (Rule 2)
per-cell scores	escape one-stop, or UCell / AUCell alone	scores describe; stats go upstairs
activity	decoupleR + CollecTRI (TF) / PROGENy (pathway)	don't read TF expression directly
many lists to compare	Metascape multi-list	overkill for a single list

Scenario 1: the ifnb stimulation response

B13's data with this episode's selection logic — same as exercise Basics-03

- 1. Scout** paste the DE list into Enrichr for direction — expect interferon themes
- 2. Fix the route** a full pseudobulk ranking exists, so the GSEA route (Rule 1)
- 3. Pick the library** Hallmark 50 first; add Reactome when mechanism details matter
- 4. Run** fgsea + msigdb; known-answer test: interferon response must top the board
- 5. Record** MSigDB version, method, ranking source — into the notes (Rule 4)

Scenario 1, minimal skeleton

Same data as B13 — and the same answer

```
ranks <- setNames(res$stat, rownames(res)) # pseudobulk 排名
hallmark <- split(hm$gene_symbols, hm$gs_name) # msigdb 抓的 H 館

library(fgsea)
set.seed(1234)
gsea <- fgsea(pathways = hallmark, stats = ranks,
              minSize = 15, maxSize = 500, eps = 0)
head(gsea[order(gsea$padj),
             c("pathway", "NES", "padj")], 2)
```

	pathway	NES	padj
1	HALLMARK_INTERFERON_ALPHA_RESPONSE	2.88	4.1e-23
2	HALLMARK_INTERFERON_GAMMA_RESPONSE	2.79	4.1e-23

On IFN-beta-stimulated data, interferon response must lead — known answer, confirmed

Scenario 2: scoring tumour states

The B16 / B19 setting — same as Advanced 03, 06, 12

- 1. Fix the route** the question is per-malignant-cell state, so the scoring route
- 2. Pick gene sets** Hallmark for direction; tumour states from the 3CA meta-programs (annotation library)
- 3. Pick the tool** escape one-stop (four scorers side by side), or UCell alone
- 4. Rule out artefacts** dissociation stress and hypoxia must clear first (B13's strategy room)
- 5. Converge** scores describe — between-group claims return to patient-level stats (Rule 2)

Scenario 2's three landmines

Circularity



scoring the data with
its own markers — the
conclusion was baked
into the input (B13,
trap 2)

Stress read as biology



clusters high on
stress / hypoxia scores:
ask about sampling
before states

Testing raw scores



t-testing thousands of
per-cell scores
resurrects
pseudoreplication —
the unit is the patient

When to leave the shortlist for the extended library

Thirteen extended resources, each with its trigger — five picked here

Your situation	Use
several clusters' marker lists to compare at once	Metascape
web-based GSEA (beyond ORA), or metabolomics	WebGestalt
PPI network plus cross-library enrichment in one query	STRING (CC BY, commercial-safe)
GO results exploding with redundancy	rrvgo / REVIGO
an all-Python pipeline	GSEAPy + decoupler-py

The whole map on one page

Segment	One line
Four iron rules	input decides method; chase the unit upstream; set the universe; record versions
Databases	open with Hallmark + GO BP; Reactome for mechanism; avoid KEGG commercially
Toolchains	test with clusterProfiler / fgsea; score with escape / UCell; infer with decoupleR
Decision tree	start from what you hold; all five branches have a home

Next steps

The reference library



the enrichment
library's three files:
overview (the tree), 16
shortlisted, 13
extended — the
source of every
number here

Matching exercises



Basics-03 (first
enrichment); Basics-06
/ Advanced-24
(pseudobulk to GSEA);
Advanced 03 / 06 / 12
(scoring)

Activity in practice



Advanced 04 / 31
(CollecTRI TF activity);
Advanced 24 / 50
(PROGENy pathway
activity)

IN ONE LINE

**A gene list is not the destination:
the right library and the right route
turn it into biology.**

And every 'right' choice — method, background, version — can be written down and traced back.

END OF THE SUPPLEMENTS

The BS series closes: all five maps are yours

Data, platforms, tools, annotation, enrichment — five supplements, one message: resources age, principles of choice do not.

The reference libraries and the exercise bank are there whenever you need them.

scRNA-seq video series • Series B Hands-on